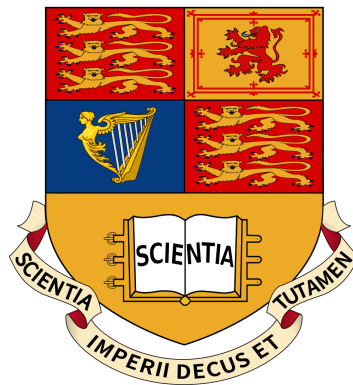


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Multimode Brillouin scattering and its temperature dependence in optical fibre

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a statement about its contents.

Abstract

Stimulated Brillouin scattering in optical fibre is commonly described by a single resonance, but the acoustic structure of the fibre can produce several closely spaced peaks associated with different guided acoustic modes. Since these modes sample the doped core and silica cladding differently, their frequency spacing contains information about the acoustic properties of the fibre.

We measure the multimode Brillouin spectrum of standard SMF-28 fibre using heterodyne detection and extract the frequencies and spacings of the observed resonances. We then model the fibre as a weak, graded acoustic waveguide. This provides a simple description of the observed mode spacing and shows how it depends on the properties of the acoustic guide.

We extend the model using temperature-dependent material properties to predict how the individual resonances and their spacings change with temperature. Much of the frequency shift is common to the acoustic modes and therefore cancels when neighbouring frequencies are subtracted. The remaining response is sensitive to differences between the thermo-acoustic properties of the doped core and silica cladding. The mode spacing can therefore provide information that is not contained in the conventional Brillouin shift alone.

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Finally, I would like to thank Anthony Leung, Brian Wang and Diyang Bai for generously giving their time and expertise to help with the experimental setup and measurements.

By convention sweet, by convention bitter; by convention hot, by convention cold; by
convention colour; but in reality: atoms and void

— Democritus

Typesetting

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1 Introduction

At sufficiently high powers, light in an optical fibre can interact with acoustic excitations of the glass through stimulated Brillouin scattering (SBS) [1]. In telecommunications, SBS is often a limitation because it transfers power from a forward-propagating pump into a frequency-shifted backward wave [2]. The same interaction also provides a useful sensing mechanism because the Brillouin frequency depends on the refractive index and acoustic properties of the fibre, both of which vary with temperature and strain [3, 4].

In standard optical fibre, the Brillouin response is often treated as a single resonance close to 11 GHz.¹ In practice, the fibre can support several guided acoustic modes with different radial profiles, and several of these can contribute to the measured spectrum. The result can be a series of closely spaced Brillouin peaks rather than one isolated resonance [5, 6].

The frequencies of these peaks depend on the acoustic confinement in the fibre. Core size and germania concentration, for example, affect the acoustic modes and the separation between their resonances [5–7]. The separation between neighbouring peaks depends on the radial acoustic confinement and mode structure, rather than only on the overall Brillouin frequency scale.

The temperature dependence of the conventional Brillouin frequency is well studied near room temperature [3], and its behaviour has also been measured over much wider temperature ranges [8, 9]. Near room temperature, it typically changes by around 1 MHz K^{−1}. Mode-dependent temperature and strain responses have also been investigated in fibres with several resolvable acoustic resonances, particularly for discriminating temperature and strain [10]. The relative motion of neighbouring acoustic modes in standard telecommunications fibre has been studied much less directly.

This thesis investigates that relative motion. We measure the multimode Brillouin spectrum of standard SMF-28 fibre using heterodyne detection, and then use a simple model of the acoustic guidance to describe the measured mode spacing and calculate how it changes with temperature.

1.1 Brillouin scattering in optical fibre

Stimulated Brillouin scattering was first observed in bulk media and later in optical fibre [11, 12]. In a fibre, an optical field can scatter from an acoustic excitation and produce light at a slightly different frequency. At sufficiently high optical power, the interaction becomes stimulated and the optical and acoustic fields become coupled.

In backward SBS, the forward-propagating pump and backward-propagating Stokes fields beat to produce a travelling intensity pattern. Electrostriction drives an acoustic wave from this pattern, and the resulting strain modulates the refractive index through the photoelastic effect. This scatters more pump light into the Stokes field, producing the

¹For an acoustic velocity of approximately 6 km s^{−1}, an 11 GHz longitudinal acoustic wave has a wavelength of about 0.5 μm.

positive feedback of the stimulated interaction [13]. The process is illustrated schematically in Figure 1.1.

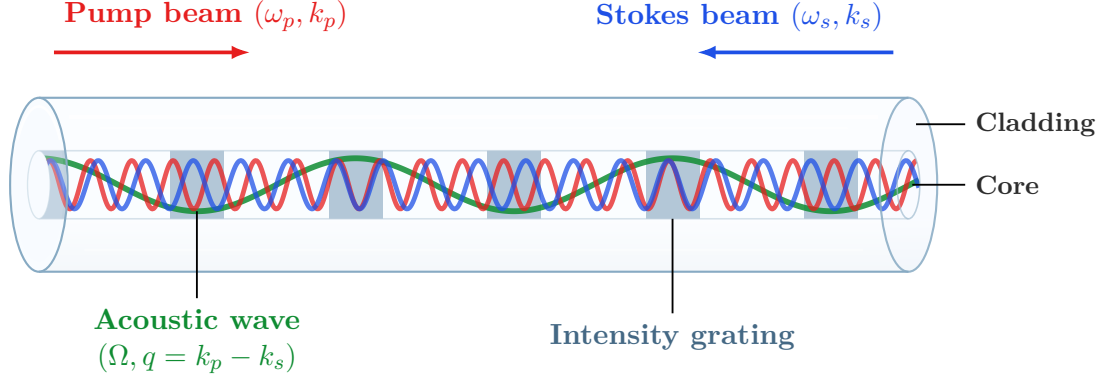


Figure 1.1: Schematic of stimulated Brillouin scattering. The counter-propagating pump and Stokes fields interact through an acoustic wave. Their beating drives the acoustic response through electrostriction, while the resulting refractive-index modulation scatters pump light into the Stokes field through the photoelastic effect.

The interaction satisfies energy and momentum conservation. For a pump field with angular frequency ω_p and wavevector k_p , scattering into a Stokes field with frequency ω_s and wavevector k_s involves an acoustic wave with angular frequency Ω and axial wavevector q , such that

$$\omega_p = \omega_s + \Omega, \quad k_p = k_s + q. \quad (1.1)$$

For backward SBS, the Stokes field travels in the opposite direction to the pump, so the two optical wavevectors are approximately equal in magnitude and opposite in direction. For a guided optical mode with effective refractive index n_{eff} and vacuum wavelength λ_p , the acoustic wavevector selected by phase matching is approximately

$$q_B = \frac{4\pi n_{\text{eff}}}{\lambda_p}. \quad (1.2)$$

For an acoustic wave with phase velocity V , the corresponding Brillouin frequency is

$$f_B = \frac{q_B V}{2\pi} = \frac{2n_{\text{eff}} V}{\lambda_p}. \quad (1.3)$$

For silica fibre at a wavelength of 1550 nm, this gives a Brillouin frequency close to 11 GHz.

At fixed pump wavelength, changes in either n_{eff} or V shift the Brillouin frequency. Since both quantities depend on temperature and strain, this dependence forms the basis of distributed Brillouin temperature and strain sensing [14]. For standard fibre at 1550 nm, a room-temperature coefficient of $1.10(2) \text{ MHz K}^{-1}$ has been measured [4].

The acoustic contribution is particularly important for the temperature response. Measurements on germanosilicate fibres show that most of the Brillouin temperature shift arises from the temperature dependence of the elastic properties of the glass rather than from the thermo-optic response alone [15]. In germanosilicate fibre, the acoustic velocity makes the dominant contribution to the temperature dependence of the Brillouin frequency, with a smaller contribution from the changing refractive index.

In the low-gain limit, an isolated Brillouin resonance is approximately Lorentzian [5]. This single-resonance description is sufficient for many conventional Brillouin measurements, but it does not account for the presence of several guided acoustic modes with different radial profiles.

1.2 Acoustic guiding and multimode Brillouin scattering

The core of a standard telecommunications fibre is doped with germania (GeO_2) to increase its refractive index and guide the optical field. Germania also reduces the longitudinal acoustic velocity of silica [6, 16], so the doped central region can act as an acoustic guide as well as an optical guide. Because the longitudinal acoustic velocity is lower in the doped central region, acoustic waves can be confined there by the surrounding higher-velocity silica. Solving the elastic wave problem gives a discrete set of acoustic modes with different radial field distributions rather than a single uniform acoustic wave [7]. These radial modes arise from the acoustic structure of the fibre itself.

The modes of interest here are the axisymmetric, predominantly longitudinal L_{0m} family, where m labels the successive radial modes. In a complete elastic description the displacement also contains a shear component. For the small material contrasts found in germanosilicate fibre, however, the modes remain predominantly longitudinal and their resonance frequencies are close to those obtained from an idealised longitudinal description [7]. A full elastic treatment also gives weak leakage, but this is expected to affect the attenuation more strongly than the resonance frequencies.

The optical phase-matching condition fixes q_B , while several radial acoustic modes can exist at this same axial wavevector. Each has a different eigenfrequency Ω_m , producing a distinct Brillouin resonance at

$$f_m = \frac{\Omega_m}{2\pi}. \quad (1.4)$$

Figure 1.2 illustrates this as several acoustic dispersion branches intersecting the same phase-matched wavevector q_B .

The modes also have different radial profiles, as illustrated in Fig. 1.3. Because the modes extend over different radial regions, changes in the material profile can shift their frequencies by different amounts. Mode-dependent overlap with the fibre composition has also been found in previous modelling of multimode Brillouin scattering [6]. The strength of each Brillouin peak depends on the overlap between the optical field and the acoustic displacement field, so not every supported acoustic mode is expected to appear equally strongly in the measured spectrum [1, 5]. The observed number and relative strength of the peaks need not match the complete set of acoustic solutions supported by the fibre.

The separation between the resonances depends on the radial acoustic confinement as well as on the overall Brillouin frequency scale. Changes in the core size, germania concentration or radial composition profile can alter the acoustic confinement and hence the separation between successive resonances [5–7].

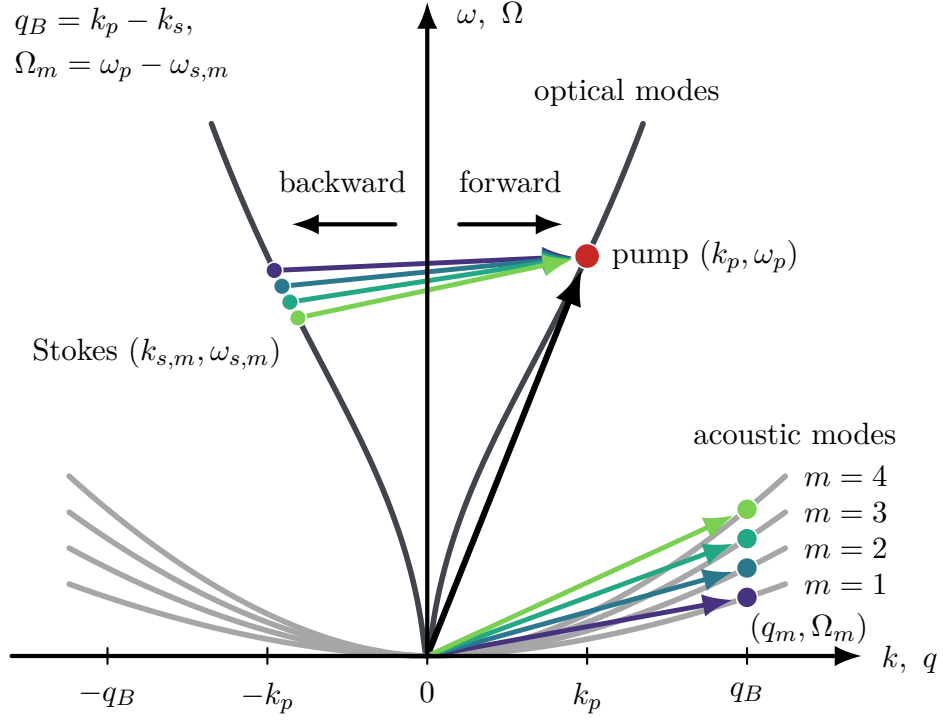


Figure 1.2: Dispersion diagram for multimode Brillouin scattering. The optical phase-matching condition selects the Brillouin wavevector q_B , at which different acoustic-mode branches have frequencies Ω_m . Each branch therefore gives a distinct Brillouin resonance at $f_m = \Omega_m/(2\pi)$.

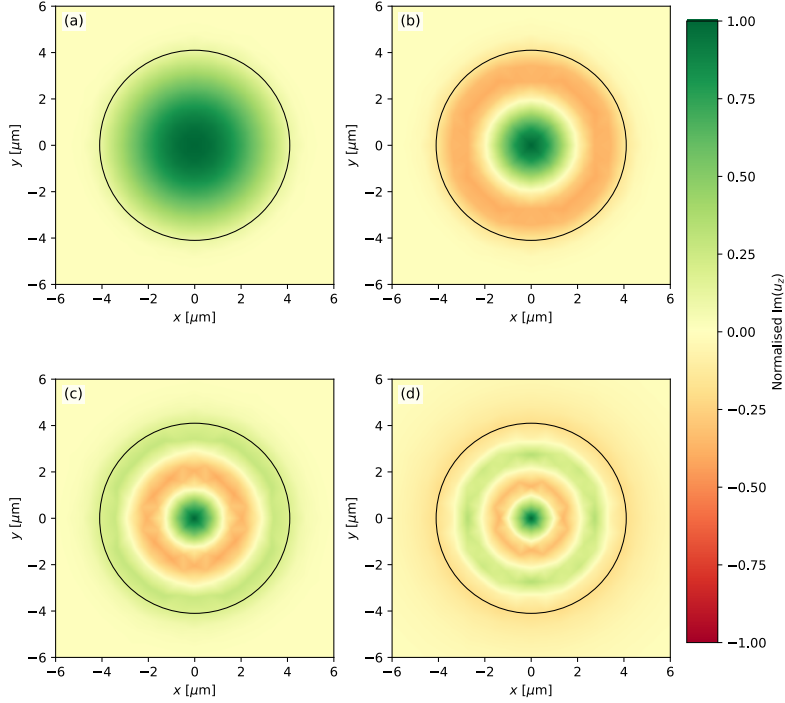


Figure 1.3: Radial displacement profiles of four axisymmetric acoustic modes calculated using NumBAT. The modes have different spatial extents and therefore sample the radial acoustic properties of the fibre differently. Higher-order modes extend further towards the core-cladding transition.

1.3 Mode spacing and temperature dependence

The spacing between two neighbouring acoustic-mode resonances is defined as

$$\Delta f_m = f_{m+1} - f_m. \quad (1.5)$$

The frequencies f_m describe the absolute position of the Brillouin spectrum, while Δf_m describes the much smaller separation between neighbouring modes. Temperature shifts the Brillouin resonances through changes in refractive index, acoustic velocity and the dimensions of the fibre. Near room temperature, the conventional Brillouin frequency changes by about 1.1 MHz K^{-1} [3, 4]. Measurements over wider temperature ranges show that the response is not simply linear at lower temperatures [8, 9]. Most Brillouin temperature measurements focus on a single resonance, although mode-resolved studies have shown that different acoustic modes can have measurably different temperature and strain responses [10, 15].

Neighbouring modes share much of the same overall temperature response, but their different radial profiles make their frequency separation sensitive to changes in the acoustic confinement. The spacing can therefore retain information about changes in the core-cladding acoustic contrast or radial guiding profile even when the much larger common Brillouin shift largely cancels.

In this work, the quantity of interest is the change in the neighbouring-mode spacing Δf_m in standard SMF-28 fibre. The conventional temperature coefficient describes the overall shift of the spectrum, while $d\Delta f_m/dT$ describes the much smaller relative motion of neighbouring modes. The analytical model developed later is used to estimate the size of this response and identify the material properties that control it.

1.4 Aims and scope

The project first measures the multimode Brillouin spectrum of standard SMF-28 fibre. We use heterodyne detection to resolve the closely spaced L_{0m} resonances and measure their centre frequencies, linewidths and neighbouring spacings. We then use a simple model of acoustic guidance to understand the frequency scale of the measured spacing. The germania-doped region is treated as a weak acoustic waveguide, relating the spacing to the longitudinal acoustic velocity, the core-cladding velocity contrast and an effective radial size of the guiding region. The model is intended to identify how these quantities set the spacing rather than to reconstruct the exact radial material profile of the fibre. The model is then extended using temperature-dependent material properties to calculate both the shifts of the individual acoustic modes and the much smaller changes in their neighbouring spacings.

Only the room-temperature multimode spectrum is measured experimentally in this project. The temperature dependence presented later is a theoretical prediction. We use published measurements of the conventional Brillouin frequency to check the overall temperature response of the model, while the predicted change in neighbouring-mode

spacing provides a target for a future temperature-dependent measurement.

2 Measuring the multimode Brillouin spectrum

The acoustic-mode structure discussed in Chapter 1 occurs on a much smaller frequency scale than the overall Brillouin shift. In SMF-28 fibre, the Brillouin frequency is close to 11 GHz, while neighbouring guided acoustic modes are separated by only a few tens of megahertz. We use heterodyne detection to resolve the individual resonances while preserving their relative frequency scale across the full multimode spectrum.

2.1 Heterodyne measurement

The experimental setup is shown in Fig. 2.1. We use a narrow-linewidth tunable laser at 1550 nm as the pump. The light is attenuated and sent through an optical circulator into 10 km of SMF-28 fibre. Brillouin-scattered light travelling backwards through the fibre returns to the circulator and is directed into the detection system. We mix the scattered Stokes light with a second tunable laser acting as a local oscillator (LO). The LO frequency is chosen so that its difference from the Stokes frequency is only a few hundred megahertz, rather than the full Brillouin shift of approximately 11 GHz. The optical frequency difference is transferred to this much lower electrical frequency while retaining the spectral structure of the Brillouin signal. This allows neighbouring resonances separated by only a few tens of megahertz to be resolved using an electrical spectrum analyser (ESA).

The Stokes and LO fields are combined on a fibre coupler and detected using a balanced photodetector. Their interference produces an electrical beat signal at the difference between the two optical frequencies. Writing the Stokes and LO frequencies as f_{Stokes} and f_{LO} , the measured beat frequency is

$$f_{\text{beat}} = |f_{\text{LO}} - f_{\text{Stokes}}|. \quad (2.1)$$

The ESA measures this heterodyne beat spectrum rather than the 11 GHz Brillouin frequency shift directly. We record the relative laser-lock frequency for each measurement, allowing the ESA frequency axis to be converted back to a Brillouin-frequency axis. For the measurements used here, the LO is locked below the Stokes frequency, so that

$$f = f_{\text{lock}} + f_{\text{beat}}. \quad (2.2)$$

This allows resonances from different measurements to be placed on the same Brillouin-frequency axis. The frequency lock has an uncertainty of approximately 2 MHz. This limits the accuracy of the absolute Brillouin frequencies, but has less effect on the spacing between neighbouring modes because a frequency offset common to both resonances cancels when their difference is taken. Relative frequency offsets between separate acquisitions

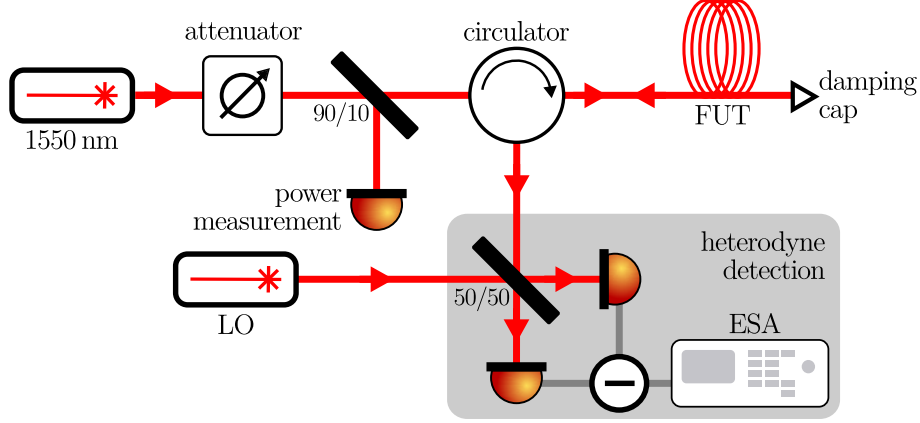


Figure 2.1: Heterodyne setup used to measure the multimode Brillouin spectrum. Backscattered light from the fibre under test (FUT) is combined with an offset local oscillator (LO) and detected on a balanced photodetector. The resulting beat signal is recorded on an electrical spectrum analyser (ESA). A separate photodetector monitors the input pump power.

do not cancel in this way and are treated below.

All measurements used in the main analysis were recorded at ambient laboratory temperature. The temperature dependence discussed later was not measured experimentally in this project, but is predicted by the acoustic model.

2.2 Spectrum reconstruction

We record the multimode Brillouin spectrum in three overlapping ESA windows so that sufficient frequency resolution can be maintained across the full spectrum. Each window is converted to the same Brillouin-frequency axis using Eq. (2.2), and the three measurements are then combined using their overlap regions.

The overlaps allow the measured powers and the positions of common spectral features to be compared between neighbouring acquisitions. We correct small amplitude differences using the common frequency regions, while the common spectral features show the relative frequency alignment between neighbouring acquisitions. Background traces are recorded separately and subtracted from the measured spectra before the amplitude correction is applied. The overlaps also show small relative frequency offsets between acquisitions. There is no independent reference for deciding which frequency axis is more accurate, so we do not shift the spectra to force agreement. Applying such a correction would make the overlap appear cleaner without establishing that the corrected axis was closer to the true frequency. Instead, we retain the observed offsets as a systematic contribution to the uncertainties of the fitted mode frequencies and spacings. Further details of the acquisition, background subtraction and reconstruction are given in Appendix A.1.

The reconstructed spectrum contains five clear Brillouin resonances between approximately 10.82 GHz and 11.14 GHz. We associate these with the first five observed axisymmetric longitudinal acoustic modes, labelled L_{01} to L_{05} .

2.3 Peak fitting and measured spectrum

The resonances overlap strongly enough that their centre frequencies cannot be found reliably from the positions of the local maxima alone. The position of a maximum is affected by the tails of the neighbouring resonances, so simply reading the peaks from the spectrum would not in general give the centre frequency of each underlying mode. We fit all five resonances simultaneously, so that the centre, width and amplitude of each mode are obtained while accounting for the overlapping tails of its neighbours. This is particularly important for the lower-frequency resonances, where the overlap is strongest.

At sufficiently low pump powers, where pump depletion and saturation can be neglected, the Brillouin response of an acoustic mode is approximately Lorentzian [5]. We fit the measured spectrum as a sum of five Lorentzians,

$$P(f) = P_0 + \sum_{m=1}^N \frac{A_m}{1 + 4(f - f_m)^2/\Gamma_m^2}, \quad (2.3)$$

where P_0 is a constant background, A_m is the fitted amplitude, f_m is the centre frequency and Γ_m is the FWHM linewidth of the m th resonance. Here, $N = 5$. The relation between this linear fit and the more general gain-dependent Brillouin response is discussed in Appendix A.2. The later analysis uses the fitted centre frequencies f_m . The amplitudes depend on the strength of the optical–acoustic interaction and the experimental conditions, while the linewidths are more sensitive to acoustic damping and the detailed gain regime.

The Lorentzian model describes the higher-frequency modes well. The L_{01} resonance deviates more noticeably from the fitted line shape. This may indicate that this part of the measurement is not fully within the low-gain regime. Since the centre frequency is the quantity required here, we retain the same Lorentzian form for all five modes. A second spectrum recorded closer to the SBS threshold and fitted using the gain-dependent form is shown in Appendix A.3.

Figure 2.2 shows the fitted spectrum together with the individual Lorentzian components and their sum. The fitted centre frequencies and linewidths are listed in Table 2.1. The spacing between neighbouring acoustic-mode resonances is defined as

$$\Delta f_m = f_{m+1} - f_m. \quad (2.4)$$

The measured neighbouring-mode spacings are 76.26 MHz, 77.63 MHz, 78.18 MHz and 81.59 MHz. The first three lie between 76.26 MHz and 78.18 MHz, giving a characteristic spacing of roughly 77 MHz. The fourth spacing is larger, at 81.59 MHz.

Although the first three spacings increase with mode number, the variation is comparable with that seen between independent measurements, so we do not treat it as clear evidence of a systematic trend. The individual spacings are compared with the acoustic model in Chapter 4. The uncertainties in the measured spacings range from 0.47 MHz to 1.11 MHz. The absolute laser-lock uncertainty is largely common to neighbouring modes and mostly cancels in their frequency difference. The remaining uncertainty comes mainly from the spectral fit and from the relative frequency alignment of the three acquisition

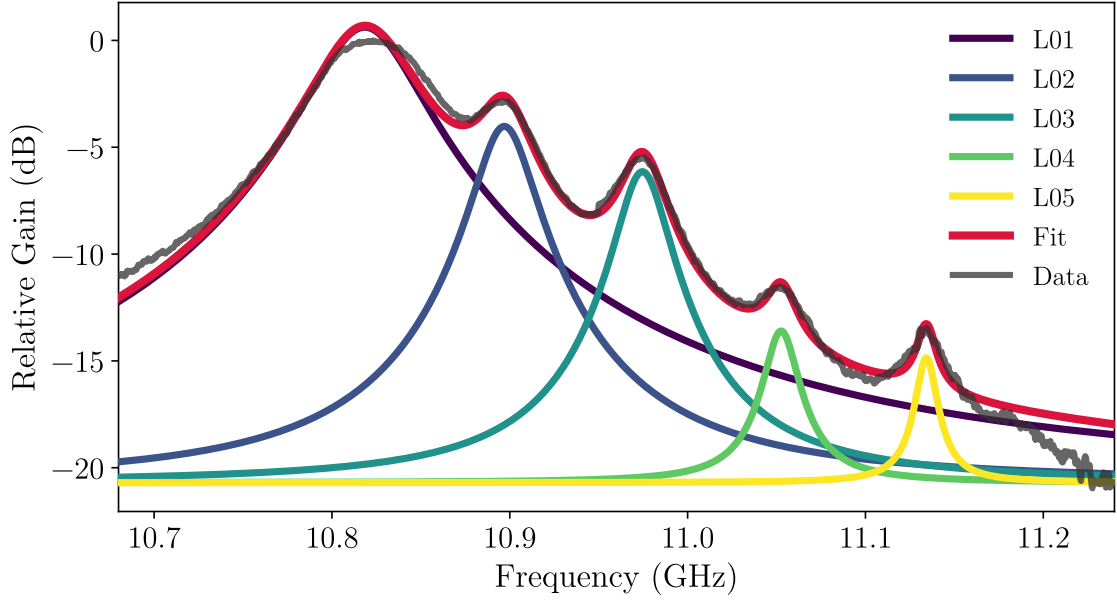


Figure 2.2: Measured multimode Brillouin spectrum and simultaneous five-mode Lorentzian fit. The individual fitted components correspond to the observed L_{0m} acoustic modes, and their sum gives the total fitted spectrum.

Table 2.1: Fitted centre frequencies f_m , FWHM linewidths Γ_m , and neighbouring-mode spacings $\Delta f_m = f_{m+1} - f_m$ for the five resolved Brillouin resonances. The quoted uncertainties σ apply to the neighbouring-mode spacings.

Mode	f_m (GHz)	Γ_m (MHz)	Δf_m (MHz)	σ (MHz)
L_{01}	10.820 68	39.5	76.26	0.51
L_{02}	10.896 94	34.2	77.63	1.11
L_{03}	10.974 57	29.7	78.18	0.47
L_{04}	11.052 75	21.8	81.59	0.52
L_{05}	11.134 35	15.6	—	—

windows. The uncertainty procedure is described in Appendix A.4. For the first spacing in particular, reasonable changes to the fitting model and fit window change the fitted value by approximately 2 MHz to 3 MHz. This additional variation is not included in the quoted uncertainty.

The fitted linewidths decrease from about 40 MHz for L_{01} to about 16 MHz for L_{05} . The poorer Lorentzian fit to L_{01} makes its fitted linewidth less reliable than those of the other modes. Since the work that follows concerns the mode frequencies rather than their damping, we do not analyse the origin of the linewidth variation further.

2.4 Experimental checks and limitations

An earlier spectrum of the same fibre gives an independent check on the measured mode-spacing scale. It was recorded under different experimental conditions and is treated as a consistency check rather than a repeat of the final measurement.

The first two neighbouring-mode spacings in the earlier spectrum are 75.56 MHz and

75.31 MHz, close to the corresponding scale found in the final dataset. They do not show the same small increase as the first two spacings of the final measurement. The agreement in the overall frequency scale appears more robust than the detailed mode-to-mode pattern. The earlier measurement is shown in Appendix A.3.

The main experimental limitation is the reconstruction of the spectrum from three separate ESA windows. Small relative frequency offsets between the windows contribute directly to the uncertainty of the measured spacings, particularly when two neighbouring resonances are determined mainly from different acquisitions. Recording the complete multimode spectrum in a single sufficiently resolved window would remove this source of relative frequency uncertainty.

The strong overlap between neighbouring resonances also introduces correlations between fitted parameters, particularly for the lower-frequency L_{0m} resonances. This makes the detailed fitted linewidths and amplitudes less secure than the centre frequencies. For the present analysis, the centre frequencies and their differences remain the quantities of primary interest.

3 An analytical model of the guided acoustic modes

The measured Brillouin modes are separated by roughly 76 MHz. We model the germania-doped central region of the fibre as a weak acoustic waveguide and use this description to calculate the mode spacing and its temperature dependence.

A parabolic acoustic-velocity profile gives a simple expression for the neighbouring-mode spacing. We then solve the same scalar wave equation numerically for the finite guide when calculating the mode frequencies and their temperature dependence. The model is intended to identify the main quantities which set the spacing and how they respond to temperature, rather than reproduce the complete acoustic structure of the fibre.

3.1 The acoustic waveguide

Germania doping increases the optical refractive index of silica but also reduces its longitudinal acoustic velocity [6, 16]. The doped core region can therefore provide weak acoustic confinement as well as optical confinement, supporting predominantly longitudinal, axisymmetric acoustic modes [7, 17].

In a complete elastic description, the displacement contains both longitudinal and shear components. These components are coupled by the boundary conditions whenever the density and shear velocity vary across the fibre. The longitudinal and shear problems decouple exactly when the density and shear velocity are the same in the core and cladding [7]. Real germanosilicate fibre does not satisfy these conditions exactly, so the modes are predominantly longitudinal and weakly leaky rather than perfectly confined. In

the weak-guidance limit, this leakage affects the attenuation more strongly than the resonance frequencies, which remain close to those predicted by the corresponding idealised guided-mode model [7]. Since the present work concerns resonance frequencies rather than acoustic damping, we use a scalar longitudinal model as a reduced description.

We assume harmonic propagation along the fibre and write the acoustic field as

$$\psi(r, z, t) = \psi(r) \exp[i(q_B z - \Omega t)],$$

where $\psi(r)$ describes the radial field, Ω is the acoustic angular frequency and q_B is the axial acoustic wavenumber. This gives

$$\frac{d^2\psi}{dr^2} + \frac{1}{r} \frac{d\psi}{dr} + \left[\frac{\Omega^2}{V_L^2(r)} - q_B^2 \right] \psi = 0, \quad (3.1)$$

where $V_L(r)$ is the longitudinal acoustic velocity as a function of radius [7]. For backward Brillouin scattering, q_B is fixed by the optical phase-matching condition in Eq. (1.2). The axial wavenumber and radial material profile are therefore known, and Eq. (3.1) is solved for the allowed acoustic frequencies Ω_m and radial fields $\psi_m(r)$. We require the radial field to remain finite at $r = 0$ and to fall away outside the guiding region. Each allowed radial solution gives a different acoustic frequency Ω_m at the same q_B , with

$$f_m = \frac{\Omega_m}{2\pi}.$$

These different radial eigenvalues give the multiple acoustic-mode resonances measured in Chapter 2. Equation (3.1) has the same mathematical structure as the scalar optical-fibre wave equation: $\Omega/V_L(r)$ plays the role of the local optical wavenumber and q_B the propagation constant [17]. Here the guiding quantity is the longitudinal acoustic velocity rather than the refractive index. A lower acoustic velocity near the fibre axis provides radial confinement, and the allowed frequencies follow from the corresponding radial eigenvalue problem.

The exact radial dopant profile of the SMF-28 fibre used in the experiment is not known, so we assume a parabolic profile. To do this, we represent the acoustic velocity using the power-law profile [18].

$$V_L(r) = \begin{cases} V_1 \left[1 + \delta \left(\frac{r}{R_g} \right)^p \right], & r \leq R_g, \\ V_2, & r > R_g, \end{cases} \quad \delta = \frac{V_2}{V_1} - 1. \quad (3.2)$$

Here V_1 is the longitudinal acoustic velocity on the fibre axis, V_2 is the cladding velocity, R_g is an effective acoustic guiding radius, p is the profile exponent, and δ is the fractional core-cladding acoustic velocity contrast. The choice $p = 2$ gives the parabolic profile used here, while increasing p moves the profile towards a step-like form [18]. Figure 3.1 shows the assumed profile near the core. The plotted range is restricted to the guiding region so that the small velocity contrast is visible. The parabolic profile is a reduced description rather

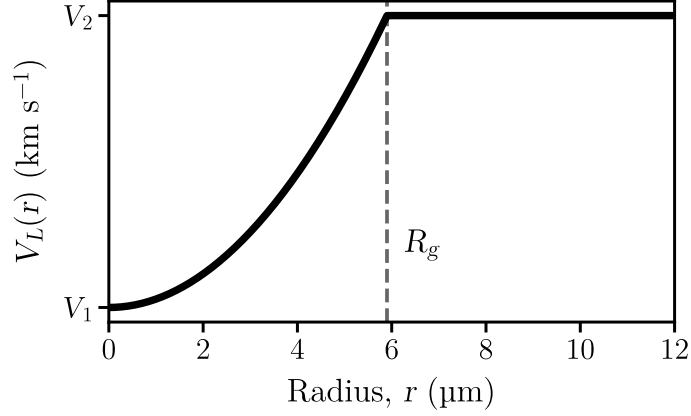


Figure 3.1: Longitudinal acoustic-velocity profile used in the model. The parabolic guiding region extends to the effective radius R_g , over which the velocity rises from the on-axis value V_1 to the cladding value V_2 . Only the region near the core is shown; the physical SMF-28 cladding extends to a radius of $62.5\text{ }\mu\text{m}$.

than an assumed measurement of the actual SMF-28 composition profile. Real fibre profiles need not be exactly parabolic or step-like, and measured commercial fibres show a range of graded and approximately uniform core profiles [5, 6]. Residual stresses introduced during fibre fabrication can also affect the relation between composition, refractive index and acoustic properties [10]. Close to the fibre axis, a smooth cylindrically symmetric profile is approximately quadratic in radius, since its slope must vanish at the centre. A parabolic profile therefore provides a simple approximation to the weak radial confinement. It also leads to approximately equal spacing between the low-order acoustic modes, similar to that observed in the measured spectrum. We fix the profile exponent at $p = 2$ and use the measured room-temperature spacing to determine the effective radius R_g . With R_g fixed in this way, the variation in spacing with mode number and its temperature dependence are predictions of the model.

We estimate the on-axis velocity from the germania concentration. Measurements of germanosilicate glass show that the longitudinal acoustic velocity decreases approximately linearly with GeO_2 content over the dilute range [16]. We write

$$V_1 = V_2 (1 - K_{V_L} C_0), \quad (3.3)$$

where C_0 is the on-axis GeO_2 concentration and K_{V_L} is the fractional reduction in longitudinal acoustic velocity per mol% of GeO_2 . Using $V_2 = 5987\text{ m s}^{-1}$, $K_{V_L} = 7.7 \times 10^{-3}$ per mol%, and $C_0 = 3.67$ mol%, with the concentration inferred for SMF-28 from its refractive-index contrast [16], gives

$$V_1 = 5817.8\text{ m s}^{-1}, \quad \delta = 0.0291.$$

The remaining room-temperature parameters are $R_g = 5.90\text{ }\mu\text{m}$ and $n_{\text{eff}} = 1.4469$ at 1550 nm , where n_{eff} is obtained from an LP_{01} calculation using the assumed room-temperature refractive-index profile.

3.2 Parabolic approximation and mode spacing

The acoustic-velocity contrast is small, with $\delta \simeq 0.029$. For the parabolic profile, Eq. (3.2) can therefore be expanded to first order in δ ,

$$\frac{\Omega^2}{V_L^2(r)} \simeq \frac{\Omega^2}{V_1^2} \left[1 - 2\delta \frac{r^2}{R_g^2} \right]. \quad (3.4)$$

Substituting this into Eq. (3.1) gives

$$\frac{d^2\psi}{dr^2} + \frac{1}{r} \frac{d\psi}{dr} + \left[\frac{\Omega^2}{V_1^2} - q_B^2 - \beta^2 r^2 \right] \psi = 0, \quad \beta = \frac{\Omega}{V_1} \frac{\sqrt{2\delta}}{R_g}, \quad (3.5)$$

where β sets the strength of the radial confinement. Equation (3.5) has the form of the radial equation for a two-dimensional harmonic oscillator with zero angular momentum. The corresponding eigenvalue condition is

$$\frac{\Omega_m^2}{V_1^2} - q_B^2 = 2\beta(2m-1), \quad m = 1, 2, \dots$$

The dependence on mode number is linear, but β itself contains Ω_m , so this relation is still implicit. Since the confinement produces only a small correction to the bulk acoustic frequency $q_B V_1$, we write

$$\Omega_m = q_B V_1 (1 + \varepsilon_m), \quad \varepsilon_m \ll 1,$$

where ε_m is the fractional frequency correction produced by the radial confinement. Keeping the leading term in ε_m gives

$$\varepsilon_m = \frac{\sqrt{2\delta}}{q_B R_g} (2m-1), \quad f_m \simeq \frac{q_B V_1}{2\pi} \left[1 + \frac{\sqrt{2\delta}}{q_B R_g} (2m-1) \right]. \quad (3.6)$$

For the room-temperature parameters, $q_B R_g \simeq 69.2$ and $\sqrt{2\delta}/(q_B R_g) \simeq 3.49 \times 10^{-3}$. Equation (3.6) then gives $\varepsilon_m \simeq 3.5 \times 10^{-3}$, 1.05×10^{-2} , 1.74×10^{-2} and 2.44×10^{-2} for the first four modes. Even for the fourth mode, the frequency correction is only about 2.4%, so treating ε_m as a small quantity remains reasonable for these modes.

The mode number appears linearly in Eq. (3.6), so neighbouring modes are equally spaced to this order. Defining $\Delta f = f_{m+1} - f_m$ gives

$$\Delta f = \frac{V_1 \sqrt{2\delta}}{\pi R_g}. \quad (3.7)$$

With $R_g = 5.90 \mu\text{m}$, this gives

$$\Delta f \simeq 75.7 \text{ MHz},$$

which reproduces the measured room-temperature spacing scale by construction.

Equation (3.7) shows how the mode spacing depends on the properties of the acoustic

guide. A larger acoustic velocity or stronger velocity contrast increases the spacing, while a larger guiding radius reduces it. The $\sqrt{\delta}$ dependence comes from the curvature of the parabolic profile.

The optical phase-matching wavenumber q_B does not appear in Eq. (3.7). In Eq. (3.6), q_B sets the main Brillouin frequency scale, but the confinement correction contains an inverse factor of q_B . These two factors cancel when neighbouring mode frequencies are subtracted. In this approximation, the absolute frequencies are therefore set by both the optical phase matching and the acoustic properties, while the spacing is determined by the radial acoustic confinement.

The measured spacing alone cannot determine V_1 , δ and R_g separately. It only fixes the combination $V_1\sqrt{2\delta}/R_g$. We therefore estimate V_1 and δ from the germania concentration and published acoustic data, and use the measured spacing to determine R_g . The value obtained for R_g consequently depends on the material parameters assumed in the model.

3.3 Finite-guide correction

The analytical result makes two main approximations. First, the acoustic-velocity contrast is expanded only to first order. Second, the harmonic-oscillator solution treats the parabolic profile as though it extends indefinitely, while the model profile reaches the cladding velocity at R_g and is constant outside this radius. To check the size of these approximations, we solve the original scalar equation, Eq. (3.1), numerically using the finite profile in Eq. (3.2) and the same boundary conditions described above. The solutions are ordered by radial mode number and converted to Brillouin frequencies using $f_m = \Omega_m/(2\pi)$.

At room temperature, the first three calculated spacings are

$$\Delta f_{12} = 75.631 \text{ MHz}, \quad \Delta f_{23} = 75.539 \text{ MHz}, \quad \Delta f_{34} = 74.775 \text{ MHz}, \quad (3.8)$$

where $\Delta f_{ij} = f_j - f_i$ denotes the separation between modes i and j . These differ from the 75.70 MHz analytical value by less than 0.3 % for the first two spacings and by about 1.2 % for the third. The leading-order expression is therefore very accurate for the lowest spacings, with the finite-guide correction becoming more noticeable for the higher-order modes.

The finite guide also introduces a weak dependence on mode number. In the analytical oscillator problem, the quadratic profile extends indefinitely and gives constant spacing at the order retained. In the finite guide, higher-order radial fields extend further towards the transition to the uniform cladding and are more strongly affected by the finite extent of the guide. The calculated spacing therefore decreases slightly with mode number. We use the finite numerical solution for the temperature-dependent calculations below. The analytical expression is still useful because it shows directly which material parameters control the spacing.

A finite weak guide supports only a limited range of confined solutions. In the parabolic approximation, confinement requires $\varepsilon_m < \delta$. For the room-temperature parameters this

predicts four confined modes. The precise cutoff depends on the assumed velocity profile and the effective value of R_g , while five spectral features are observed experimentally. The fifth feature is not represented as a confined mode in the present simplified model. This does not by itself determine which modes should be visible experimentally, since their spectral strengths also depend on optical–acoustic overlap.

3.4 Temperature-dependent material properties

To extend the model with temperature, we need $V_1(T)$, $V_2(T)$, $R_g(T)$ and $n_{\text{eff}}(T)$. The first three determine the mode spacing, while $n_{\text{eff}}(T)$ enters the absolute frequencies through qB .

We take the cladding velocity from published measurements of silica and relate the core velocity to the cladding response using measurements of the composition dependence of germanosilicate fibre. The refractive index follows the measured thermo-optic response, while the effective acoustic radius is assumed to follow the thermal expansion of the glass.

3.4.1 Cladding velocity

The longitudinal acoustic velocity of silica is not monotonic with temperature. On cooling from room temperature it initially decreases, reaches a minimum, and then rises again at lower temperatures [19, 20]. This behaviour produces the broad minimum in the Brillouin frequency seen in cryogenic measurements.

We represent the silica velocity as the sum of a relaxational contribution and a smooth background [20],

$$V_L(T, \Omega) = W(T) + v_0 \left(\frac{\delta v}{v} \right)_{\text{rel}} (T, \Omega), \quad (3.9)$$

where $W(T)$ is the smooth background velocity, v_0 is the velocity scale used in the relaxational model, and $(\delta v/v)_{\text{rel}}$ is the fractional relaxational correction. We take the relaxational term and its parameters from Ref. [20]. Since this contribution depends on frequency, we evaluate it at the acoustic frequency near 11 GHz relevant to the present experiment.

The smooth background is written as

$$W(T) = W_0 \left[1 - \frac{a}{e^{\Theta/T} - 1} \right], \quad (3.10)$$

where W_0 sets the velocity scale, a determines the size of the temperature correction, and Θ gives its characteristic temperature. We use this as a smooth empirical description of the unrelaxed acoustic response of vitreous silica [20–22]. We constrain the background using bulk-silica Brillouin measurements over the low-temperature range [19], together with the measured room-temperature Brillouin temperature coefficient at 1.55 μm [4]. The low-temperature measurements end well below room temperature and do not by themselves determine the slope at 298.15 K, so the room-temperature coefficient provides a second constraint. At 298.15 K, the resulting fractional temperature coefficient of the cladding

longitudinal velocity is

$$\xi_2 \equiv \frac{d \ln V_2}{dT} = 9.74 \times 10^{-5} \text{ K}^{-1}. \quad (3.11)$$

We normalise the absolute velocity to $V_2 = 5987 \text{ m s}^{-1}$ at 298.15 K. The corresponding room-temperature slope is consistent with the reported value $dV_L/dT = 0.57 \text{ m s}^{-1} \text{ K}^{-1}$ [16, 23]. This provides a consistency check between the literature values rather than an independent test of the model.

The three-parameter background is a smooth empirical parameterisation rather than an exact description of the silica data. Its residuals against the bulk-silica measurements are systematic at the level of a few m s^{-1} , equivalent to several megahertz on the absolute Brillouin-frequency scale. Since this contribution is largely common to neighbouring modes, much of it cancels in the spacing.

3.4.2 Core velocity

The germania-doped core and silica cladding can have slightly different fractional temperature coefficients. This matters because the acoustic confinement depends on their velocity ratio rather than on either velocity alone. We define

$$\xi_1 = \frac{d \ln V_1}{dT}, \quad \xi_2 = \frac{d \ln V_2}{dT}, \quad \Delta\xi_{V_L} = \xi_1 - \xi_2. \quad (3.12)$$

We construct the core velocity $V_1(T)$ from $V_2(T)$ using this difference.

There is no direct measurement of $\Delta\xi_{V_L}$ for the particular SMF-28 fibre used here. We infer it from measurements of how the Brillouin temperature coefficient changes with germania concentration [15, 24]. We assume that most of this composition dependence comes from the acoustic velocity rather than the refractive index, consistent with the dominant elastic contribution reported for germanosilicate fibre [15]. The two estimates span

$$-6.9 \times 10^{-6} \text{ K}^{-1} \lesssim \Delta\xi_{V_L} \lesssim -2.7 \times 10^{-6} \text{ K}^{-1}. \quad (3.13)$$

We use the midpoint, $\Delta\xi_{V_L} = -4.8 \times 10^{-6} \text{ K}^{-1}$, as the central value. This is not a direct measurement or fitted value, and the full range is retained as a sensitivity range rather than a statistical uncertainty. Both datasets give the same sign: the fractional temperature coefficient of the germania-doped core is slightly smaller than that of the silica cladding [15, 24]. This difference changes the core-cladding velocity contrast and hence the mode spacing.

3.4.3 Refractive index and thermal expansion

The refractive index cancels from the leading-order analytical mode spacing, but it is still required for the absolute mode frequencies through q_B and is included in the finite-guide calculation.

We take the temperature dependence of the refractive index at $1.55 \mu\text{m}$ from Ref. [25]. At room temperature,

$$\frac{1}{n_{\text{eff}}} \frac{dn_{\text{eff}}}{dT} = 5.69 \times 10^{-6} \text{ K}^{-1}.$$

The data in Ref. [25] describe pure silica. We use this response for $n_{\text{eff}}(T)$ and neglect the small correction associated with the germania-doped core; the composition dependence of the thermo-optic coefficient in doped silica is discussed in Ref. [16]. We apply the silica thermo-optic variation to the room-temperature value of n_{eff} when calculating $q_B(T)$. We assume that the acoustic guide radius follows the thermal expansion of the silica cladding. We take the linear thermal expansion coefficient $\alpha(T)$ from dilatometry measurements of fused silica [26]. At room temperature it is approximately $4.5 \times 10^{-7} \text{ K}^{-1}$, much smaller than the fractional acoustic-velocity coefficients. We retain thermal expansion in the calculation, although its contribution to the spacing coefficient is expected to be small.

3.5 Temperature dependence of the mode spacing

Including the temperature-dependent material properties gives

$$f_m(T) = \frac{q_B(T)V_1(T)}{2\pi} \left[1 + \frac{\sqrt{2\delta(T)}}{q_B(T)R_g(T)}(2m-1) \right], \quad q_B(T) = \frac{4\pi n_{\text{eff}}(T)}{\lambda_p}, \quad (3.14)$$

with

$$\Delta f(T) = \frac{V_1(T)\sqrt{2\delta(T)}}{\pi R_g(T)}, \quad \delta(T) = \frac{V_2(T)}{V_1(T)} - 1. \quad (3.15)$$

If the core and cladding had the same fractional acoustic-velocity coefficient, $\xi_1 = \xi_2$, then V_1 and V_2 would scale together with temperature and δ would remain constant. In this case the core-cladding velocity contrast would stay unchanged even as the overall acoustic frequency scale shifted. When the two fractional coefficients differ, the guide contrast also changes. Differentiating $\delta = V_2/V_1 - 1$ gives

$$\frac{d\delta}{dT} = (1 + \delta) (\xi_2 - \xi_1).$$

Taking the logarithmic derivative of Eq. (3.15) gives

$$\boxed{\frac{1}{\Delta f} \frac{d\Delta f}{dT} = \xi_1 - \alpha + \frac{1 + \delta}{2\delta} (\xi_2 - \xi_1)}. \quad (3.16)$$

Equation (3.16) separates the spacing coefficient into the common temperature dependence of the core acoustic velocity, the effect of thermal expansion, and a contribution from the difference between the thermo-acoustic responses of the core and cladding.

For the room-temperature contrast used here, the prefactor $(1 + \delta)/(2\delta)$ is about 18. The small difference between the fractional core and cladding velocity coefficients is therefore strongly enhanced in the spacing coefficient. This happens because the underlying core-cladding velocity contrast is itself only $\delta \simeq 0.029$.

No thermo-optic term appears in Eq. (3.16), because q_B cancels from the neighbouring-mode spacing at this order. Changes in refractive index shift the absolute mode frequencies together without directly changing their separation.

4 Results

The absolute frequencies and neighbouring-mode spacings provide different comparisons with the model. The absolute frequencies depend mainly on the acoustic velocity and optical phase-matching condition, whereas the neighbouring-mode spacings depend on the radial acoustic confinement. Since R_g was chosen from the measured spacing scale, agreement in the overall spacing is a calibration rather than an independent prediction. The absolute frequencies and the variation of spacing with mode number were not used in this calibration.

4.1 Room-temperature spacings

At 298.15 K, the finite-guide model gives the first three neighbouring-mode spacings shown in Table 4.1. The model gives a spacing close to 75 MHz for all three mode pairs, consistent

Table 4.1: Measured and finite-guide modelled neighbouring-mode spacings at 298.15 K.

Mode spacing	Experiment (MHz)	σ (MHz)	Model (MHz)
Δf_{12}	76.26	0.51	75.63
Δf_{23}	77.63	1.11	75.54
Δf_{34}	78.18	0.47	74.78

with the overall scale of the measured spectrum. Because the effective guide radius was chosen from the room-temperature spacing, this agreement is a calibration of the model rather than an independent prediction. The differences between measured and calculated spacings are approximately 0.63 MHz, 2.09 MHz and 3.40 MHz for the first three mode pairs. The first two are comparable to the quoted measurement uncertainties, while the third is clearly larger. The measured spacings also increase from 76.26 MHz to 78.18 MHz, whereas the finite parabolic model predicts a small decrease from 75.63 MHz to 74.78 MHz.

We retain the value of R_g set by the overall spacing scale, since changing it would mainly shift all three spacings together and would not reproduce the observed trend. The parabolic model captures the characteristic spacing scale but not its detailed variation with mode number. The measured first 10.82068, 10.89694, 10.97457 and 11.05275 GHz, compared with calculated values of 10.89936, 10.97500, 11.05054 and 11.12531 GHz. The model is higher by approximately 73 MHz to 79 MHz, with a mean offset of about 76 MHz. This is only about 0.7% of the 11 GHz Brillouin frequency, but is much larger than the experimental uncertainty.

For the room-temperature parameters used here, the scalar model supports confined solutions over approximately 10.8615 GHz to 11.1773 GHz. The lowest measured resonance, at 10.82068 GHz, lies below this range and cannot be represented as a confined core-guided mode within the present parameter set. The comparison of the absolute frequencies also depends on how the measured peaks are assigned to radial modes. Shifting the correspondence by one mode, so that calculated mode m is compared with measured mode $m + 1$, reduces the first three differences to only a few MHz. This shows that the

absolute offset is not enough to distinguish between uncertainty in the material parameters and uncertainty in the mode assignment. We retain the original assignment because the simplified model supports only four confined modes while five spectral features are observed, so it does not provide a complete alternative assignment. Adjusting R_g is not an appropriate way to remove this offset. Equation (3.6) shows that the dominant absolute frequency scale is $q_B V_1 / (2\pi)$, set mainly by n_{eff} and V_1 , while R_g controls the smaller correction due to acoustic confinement.

4.2 Mode frequencies with temperature

Figure 4.1 shows the calculated frequencies of the first four acoustic modes between 5 K and 350 K. All four modes follow nearly the same temperature dependence. As the fibre is cooled from room temperature, the frequencies decrease to a minimum near 60 K and then increase again at lower temperatures. The calculated minimum occurs at 60.7 K for all four

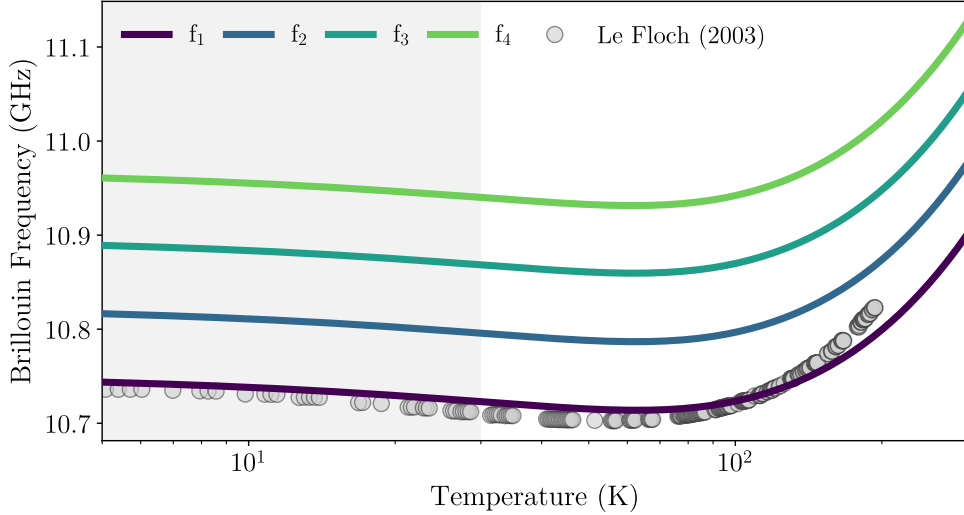


Figure 4.1: Calculated temperature dependence of the first four acoustic-mode frequencies, compared with the measurements of Ref. [8]. A constant 106 MHz offset is applied to the measured frequencies to compare their temperature dependence independently of the absolute Brillouin frequency. Temperature is plotted on a logarithmic scale, and shaded regions mark temperatures for which not all material inputs are considered reliable.

modes and follows from the non-monotonic silica acoustic velocity used in the material model, including the relaxational contribution from Ref. [20]. The measurements of Ref. [8] used a different fibre, so a direct comparison of the absolute Brillouin frequencies is not expected. We compare the temperature dependence after removing a constant frequency offset. Removing a constant offset does not change the slope, curvature or position of the minimum

The measured minimum occurs near 56 K [8], compared with 60.7 K in the model. The model reproduces the broad minimum reasonably well, although the intermediate-temperature slope is less accurate. Between 150 K and 194 K, the measured fractional temperature coefficient is approximately $1.20 \times 10^{-4} \text{ K}^{-1}$, compared with $0.81 \times 10^{-4} \text{ K}^{-1}$ in the model. The model underestimates the slope by about one third over this interval.

The measurements in Ref. [8] were obtained using different fibre lengths and experimental arrangements across the full temperature range, so this comparison should not be over-interpreted. Near room temperature, the calculated coefficients of the first four modes lie between approximately 1.07 MHz K^{-1} and 1.12 MHz K^{-1} . These values are consistent with the reported coefficient of $1.10(2) \text{ MHz K}^{-1}$ for standard fibre at $1.55 \mu\text{m}$ [4]. This agreement is expected because the room-temperature value was used to constrain the silica background in Section 3.4.1, so it is not an independent test of the model. Similar coefficients have also been measured in germanosilicate and SMF-28 fibres [15, 24]. The low-temperature minimum provides a more independent comparison, since it was not used to constrain the model

Below roughly 30 K, we do not treat the calculation as quantitative. Additional acoustic loss and velocity contributions associated with two-level systems in amorphous silica become important in this regime [9], and these are not included in the present material model.

4.3 Neighbouring-mode spacings with temperature

Figure 4.2 shows the first three neighbouring-mode spacings from the finite-guide calculation. Their small differences arise from the finite radial profile. The temperature

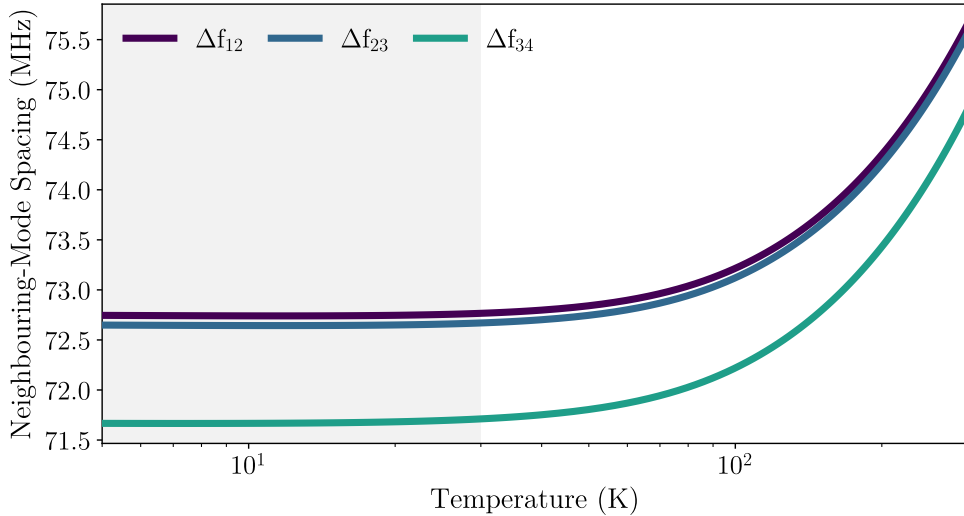


Figure 4.2: Predicted temperature dependence of the first three neighbouring-mode spacings from the finite-guide model. Temperature is plotted on a logarithmic scale, and shaded regions mark temperatures for which not all material inputs are considered reliable.

dependence of the spacing is much smaller than that of the individual frequencies. All three spacings increase as the fibre is warmed. Between 77 K and 298.15 K, the total increase is approximately 2.6 MHz to 2.8 MHz, depending on the mode pair.

The individual mode frequencies show a clear minimum near 60 K, whereas there is no corresponding minimum in the neighbouring-mode spacings. The non-monotonic silica response shifts neighbouring modes almost together and largely cancels in their frequency difference. The remaining spacing variation is mainly sensitive to the temperature dependence of the acoustic-velocity contrast between the germania-doped central region and the

silica cladding.

The predicted change from 77 K to room temperature is only a few megahertz, comparable with the present experimental spacing uncertainties. The quoted room-temperature uncertainties lie between 0.47 MHz and 1.11 MHz, and the first spacing also has an additional fit-model and fit-window sensitivity of approximately 2 MHz to 3 MHz that is not included in its quoted uncertainty. Measuring this temperature dependence would require better control of the line-shape and relative-frequency uncertainties, particularly for the first spacing.

4.3.1 Room-temperature coefficients

The calculated room-temperature spacing coefficients are listed in Table 4.2. The indi-

Table 4.2: Finite-guide predictions for the temperature coefficients of the neighbouring-mode spacings at 298.15 K.

Mode spacing	$d\Delta f/dT$ (kHz K ⁻¹)
Δf_{12}	13.39
Δf_{23}	13.40
Δf_{34}	14.02

vidual mode frequencies shift by a little over 1 MHz K⁻¹, while the spacings change by only about 13 kHz K⁻¹. Most of the temperature dependence is common to neighbouring modes and cancels in their frequency difference.

The first two coefficients are almost identical, while the third is about 5 % larger. This is consistent with the higher-order modes sampling more of the outer part of the assumed velocity profile. The exact mode-to-mode differences depend on that profile and are not treated as a robust prediction. A representative room-temperature coefficient is

$$\boxed{\frac{d\Delta f}{dT} \simeq 13.4 \text{ kHz K}^{-1}}. \quad (4.1)$$

The dominant uncertainty in the model is the differential acoustic-velocity coefficient $\Delta\xi_{V_L}$. Using the range in Eq. (3.13) gives

$$\frac{d\Delta f}{dT} \simeq 10.7 \text{ kHz K}^{-1} \text{ to } 16.0 \text{ kHz K}^{-1}.$$

This is a sensitivity range from the two literature estimates, not a statistical confidence interval. This range is much larger than the differences between the three mode-pair coefficients, but the predicted spacing coefficient remains positive throughout

5 Discussion

5.1 Comparison with the measured spectrum

The room-temperature measurements show a series of closely spaced Brillouin resonances with a characteristic separation of approximately 76 MHz. This is consistent with the picture developed in Chapter 3, in which the germania-doped central region of the fibre acts as a weak acoustic guide. In the parabolic model, the spacing is set by the longitudinal acoustic velocity, the core-cladding velocity contrast and the effective acoustic guide radius. Choosing $R_g = 5.90 \mu\text{m}$ gives the observed spacing scale.

This agreement should be interpreted with some care because R_g was chosen from the measured room-temperature spacing. The model therefore does not predict the 76 MHz scale independently. Instead, the measurement constrains the combination $V_1 \sqrt{2\delta}/R_g$, with V_1 and δ estimated from the assumed fibre composition. The fitted value of R_g is consequently best regarded as an effective length scale for the reduced acoustic model rather than as a measurement of the physical core radius.

Once the overall spacing scale has been fixed, the variation between successive mode pairs gives a more useful comparison. The first three measured spacings increase from 76.26 MHz to 78.18 MHz, whereas the finite-guide calculation decreases from 75.63 MHz to 74.78 MHz. The disagreement is largest for the higher-order pair. This is also where the model is expected to be most sensitive to the assumed radial profile, since the higher-order modes extend further from the fibre axis and place more of their displacement near the outer part of the doped region, where the parabolic profile joins the uniform cladding.

The increase seen in the final experimental dataset should not, however, be overinterpreted. The earlier independent spectrum discussed in Section 2.4 gives its first two neighbouring spacings close to 75.5 MHz and does not show the same increase with mode number. The overall spacing scale is therefore better established experimentally than the detailed sequence of the individual spacings.

If a systematic mode-dependent trend were confirmed in a future measurement, it could nevertheless provide useful information about the radial structure of the fibre. Different acoustic modes sample different parts of $V_L(r)$, so their relative frequencies depend on the shape of the acoustic guide as well as its overall strength. A departure from equal spacing could therefore indicate that the real acoustic-velocity profile differs from the simple parabolic form assumed here.

A small number of measured spacings would not be sufficient to reconstruct that profile uniquely. Different radial profiles can produce similar low-order eigenfrequencies, particularly when only three or four frequency differences are available. An independent measurement of the refractive-index or germania concentration profile would reduce this ambiguity. Since germania changes both the optical refractive index and the longitudinal acoustic velocity [6, 16], such a measurement could be used to construct a more realistic $V_L(r)$ with fewer free parameters. The measured spacing sequence could then be used as a more direct test of the acoustic model rather than partly determining it.

The absolute mode frequencies provide a separate indication of the limitations of the

reduced description. The calculated frequencies are systematically high by about 76 MHz. This is less than one percent of the approximately 11 GHz Brillouin frequency, so the overall frequency scale is captured reasonably well, but the offset is much larger than the experimental uncertainty.

Changing R_g cannot remove this common offset without also changing the mode spacing, because R_g mainly controls the transverse confinement rather than the dominant Brillouin frequency. The difference is more naturally associated with uncertainties in the quantities that set the absolute frequency scale, including the longitudinal acoustic velocity, the optical effective index and the detailed composition profile. None of these is known exactly for the particular fibre used in the experiment.

There is also some ambiguity in assigning the measured resonances to the calculated radial mode numbers. A different assignment can reduce the common frequency offset substantially. The present scalar model is not sufficiently reliable near the edge of confinement to determine the number of observable modes from first principles, and it supports only four confined modes while five spectral features are observed. For this reason, the absolute-frequency offset alone is not enough to choose between possible assignments. The neighbouring-mode spacings provide the more useful comparison with the present reduced model.

5.2 The mode spacing as a differential measurement

The temperature calculation shows a clear separation between the motion of the individual Brillouin resonances and the change in their relative positions. Near room temperature, the individual mode frequencies shift by a little over 1 MHz K^{-1} , whereas the neighbouring-mode spacing changes by only about 13 kHz K^{-1} . The same behaviour is seen over the wider temperature range. The broad minimum near 60 K is clear in the individual mode frequencies but is almost absent from their differences.

Most of the temperature response therefore moves the multimode spectrum together. The acoustic velocity of silica changes sufficiently with temperature to shift every Brillouin resonance, but neighbouring radial modes are formed from essentially the same materials and share most of this response. When one mode frequency is subtracted from the next, much of the common shift disappears.

The analytical model gives the same physical picture. The absolute frequencies contain the large common scale $q_B V_1 / (2\pi)$, while the neighbouring-mode spacing depends mainly on $V_1 \sqrt{\delta} / R_g$. The spacing is therefore more sensitive to changes in the acoustic confinement than to the overall motion of the Brillouin spectrum.

In the central model, the fractional temperature coefficient of the cladding acoustic velocity is slightly larger than that of the germania-doped central region. As the fibre is warmed, the ratio V_2/V_1 increases and the acoustic velocity contrast becomes slightly larger. This strengthens the radial acoustic guide and increases the separation between neighbouring radial modes.

Thermal expansion produces a smaller effect in the opposite direction. As the transverse scale of the guide increases, the mode separation tends to decrease. Near room

temperature this contribution is much weaker than the change in acoustic contrast, so the net prediction is an increase in the neighbouring-mode spacing with temperature.

The sensitivity to the difference between the core and cladding responses is enhanced by the weakness of the acoustic guide. For the present value of δ , the factor $(1 + \delta)/(2\delta)$ is approximately 18. A small difference between the fractional temperature coefficients of V_1 and V_2 can therefore produce a comparatively large fractional change in the guide strength, even though the two acoustic velocities themselves respond similarly to temperature.

The refractive-index response behaves differently. A change in refractive index changes the optical phase-matching wavevector and shifts the absolute Brillouin frequencies, but q_B cancels from the neighbouring-mode spacing to leading order. The spacing is consequently much less sensitive to the thermo-optic response than the conventional Brillouin frequency.

In this sense, the neighbouring-mode spacing provides a differential measurement of the acoustic response. The absolute Brillouin frequency is dominated by changes that are common across the fibre, while the spacing retains greater sensitivity to differences between the acoustic properties of the doped central region and the silica cladding.

5.2.1 What a temperature measurement could determine

In the present calculation, published composition-dependent Brillouin measurements are used to estimate $\Delta\xi_{V_L}$ and from this predict $d\Delta f/dT$. A temperature-resolved measurement of the multimode spectrum would allow the same relation to be used in the opposite direction. If the cladding response, acoustic contrast and thermal expansion were known or separately modelled, the measured spacing coefficient could provide a fibre-specific estimate of the differential core-cladding acoustic response.

Writing $\xi_1 = \xi_2 + \Delta\xi_{V_L}$ in Eq. (3.16) gives

$$\frac{1}{\Delta f} \frac{d\Delta f}{dT} = \xi_2 - \alpha + \Delta\xi_{V_L} \left[1 - \frac{1 + \delta}{2\delta} \right]. \quad (5.1)$$

For $\delta \simeq 0.029$, the coefficient multiplying $\Delta\xi_{V_L}$ is large. The spacing slope is therefore particularly sensitive to this differential response, which is also the least well constrained material input in the present calculation.

Measuring several neighbouring-mode spacings as a function of temperature would provide further information. If the different Δf_m had similar temperature coefficients, this would be consistent with the acoustic guide becoming slightly stronger or weaker while retaining approximately the same radial shape. A systematic variation with mode number would instead suggest that the thermo-acoustic response changes across the guide.

Because the different radial modes sample different parts of the fibre, such a trend could contain some information about where the change occurs. A temperature-dependent variation concentrated near the core-cladding transition, for example, would be expected to affect higher-order modes more strongly than modes concentrated closer to the fibre axis. The sign and mode-number dependence of the measured slopes could therefore help distinguish between different changes in the radial acoustic profile.

Such an interpretation would not be unique. Mode-dependent behaviour could also

arise from the finite extent of the acoustic guide, inaccuracies in the assumed radial profile, or corrections to the scalar treatment. Measurements of several mode pairs would therefore provide a stronger test of the model, even if they could not by themselves identify a unique microscopic mechanism.

5.2.2 Possible physical origins of the spacing response

The present model includes the simplest source of differential motion by allowing the longitudinal acoustic velocities of the core and cladding to have slightly different fractional temperature coefficients. Their ratio then changes with temperature, changing the depth of the acoustic guide.

The real fibre is likely to have a more gradual radial response. The germania concentration varies continuously across the doped region, and the local elastic properties vary with it. Their temperature dependence may also vary with radius. Warming the fibre could therefore change the shape of $V_L(r)$ slightly rather than simply increasing or decreasing one effective core-cladding contrast.

This would affect the different acoustic modes by different amounts. Lower-order modes place more of their displacement near the centre of the guide, while higher-order modes extend further towards the outer part of the doped region. A temperature-dependent change near the core-cladding transition could therefore shift a higher-order mode more strongly than a lower-order one and produce different temperature coefficients for the neighbouring-mode spacings.

Fabrication may introduce further deviations from the simple composition-based profile. Drawing and residual stress can modify the optical and elastic properties of the fibre [10], and these effects need not be distributed uniformly across the core. Their temperature dependence is not known for the present fibre, so there is no basis for assigning an observed spacing change specifically to residual stress. They do, however, provide another reason why the real radial acoustic profile may differ from the smooth profile used in the calculation.

The scalar approximation is another possible source of small mode-dependent differences. The real acoustic modes contain both longitudinal and shear displacement, and a complete elastic calculation can alter their detailed resonance frequencies [7]. Such corrections are expected to be small compared with the absolute Brillouin frequency, but they may be more visible in Δf_m because much of the common longitudinal frequency shift cancels when neighbouring modes are subtracted.

None of these additional effects is identified experimentally in the present work. They should therefore be regarded as possible contributions to a future mode-dependent temperature response rather than as explanations of the spectrum measured here.

5.3 Limitations

The main uncertainty in the predicted spacing coefficient is $\Delta\xi_{V_L}$, the difference between the fractional acoustic-velocity temperature coefficients of the core and cladding. This

quantity has not been measured directly for the particular SMF-28 fibre used here and is instead estimated from composition-dependent Brillouin measurements [15, 24].

The literature estimates span

$$\Delta\xi_{V_L} = -6.9 \times 10^{-6} \text{ K}^{-1} \text{ to } -2.7 \times 10^{-6} \text{ K}^{-1},$$

which gives a room-temperature spacing coefficient of approximately 10.7 kHz K^{-1} to 16.0 kHz K^{-1} . The central prediction of 13.4 kHz K^{-1} should therefore be regarded as a representative value rather than a precise prediction for the measured fibre.

The calculation also assumes that $\Delta\xi_{V_L}$ is independent of temperature. This is most defensible near room temperature, where the composition-dependent measurements are relevant. At lower temperatures, the relative elastic response of the germania-doped and undoped regions is much less well constrained. The broad behaviour of the spacing curves remains useful for understanding the suppression of the common response, but their detailed low-temperature shape should not be given the same weight as the room-temperature coefficient.

A second limitation is the unknown radial composition profile of the measured fibre. We represent the acoustic guide using a truncated parabola with an effective radius determined from the room-temperature spacing. This is enough to reproduce the characteristic spacing scale, but it limits what can be learned from the detailed differences between individual mode pairs. The disagreement between the measured and calculated spacing sequences is likely to be more sensitive to this assumption than the overall 76 MHz scale.

The scalar treatment introduces a further approximation. The real axisymmetric modes contain both longitudinal and shear displacement and are weakly leaky because the material properties do not match exactly across the guide [7]. These corrections are expected to affect attenuation more strongly than the main resonance-frequency scale, although their effect on small differences between neighbouring modes is less certain.

There is also uncertainty in the temperature-dependent material data. Much of the acoustic information used here comes from bulk vitreous silica or from fibres with different compositions and fabrication histories. Drawing conditions, residual stress and thermal history can alter both the refractive index and acoustic properties [10]. The model therefore combines measurements from several sources to construct an effective description of the particular fibre used in the experiment.

These limitations affect different conclusions to different degrees. The large common temperature shift of the acoustic modes, and its strong suppression in neighbouring-mode differences, follow mainly from the large separation between the absolute-frequency and mode-spacing scales and are the more robust features of the calculation. The precise spacing coefficient is less secure because it depends on the much smaller difference between the core and cladding acoustic responses, particularly $\Delta\xi_{V_L}$. The differences between individual mode-pair coefficients depend still more strongly on the assumed radial profile, while the detailed behaviour at very low temperature is the least well constrained part of the prediction.

5.4 Outlook

The clearest next experiment would be to measure the multimode Brillouin spectrum while changing the fibre temperature. Cooling from room temperature to around liquid-nitrogen temperature is predicted to reduce the neighbouring-mode spacings by approximately 2.6 MHz to 2.8 MHz, depending on the mode pair. This is comparable with the present spacing uncertainties, particularly the additional 2 MHz to 3 MHz fit-model and fit-window sensitivity of the first spacing. The predicted effect should therefore be accessible, but better control of the line shape and relative frequency scale would be needed.

Both the absolute frequencies $f_m(T)$ and the neighbouring differences $\Delta f_m(T)$ would be useful in such a measurement. The former would test the overall thermo-acoustic response of the model, while the latter would probe changes in the radial acoustic confinement. Measurements at several temperatures would allow both the local slope and any curvature in $\Delta f_m(T)$ to be determined, and following several mode pairs would show whether the different radial modes respond in the same way. A room-temperature spacing slope near the predicted $\sim 13 \text{ kHz K}^{-1}$ scale would support the central value of $\Delta \xi_{V_L}$, while a different measured slope could be used with Eq. (5.1) to estimate $\Delta \xi_{V_L}$ for the particular fibre.

The experimental procedure could also be improved by recording the complete multimode spectrum in a single ESA window. This would remove the relative frequency offsets introduced when separate spectral windows are stitched together. For a spacing measurement, stability of the relative frequency scale between neighbouring resonances is more important than very accurate knowledge of the absolute Brillouin frequency.

A further improvement would be an independent measurement of the radial refractive-index or composition profile. Since germania changes both the refractive index and acoustic velocity, such a profile could be used to construct a more realistic estimate of $V_L(r)$. The effective guide radius would then no longer need to be chosen from the measured spacing, allowing the room-temperature modal spectrum to become a genuine prediction of the model. Combined with temperature-dependent measurements, this would make it possible to test whether the mode-to-mode behaviour follows from the composition profile alone or whether additional elastic effects are needed.

The experiment could also be extended to lower temperatures, where additional relaxation and two-level-system effects become important in amorphous silica [9]. The present model is not intended to describe this regime quantitatively, but a multimode measurement could show whether the additional low-temperature acoustic response is shared by the different modes or whether it also changes their relative spacing.

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Appendix A

Additional experimental details

This appendix gives further details of the spectrum reconstruction, gain-dependent line shape and frequency uncertainties used in Chapter 2.

A.1 Spectrum acquisition and reconstruction

We assemble the final room-temperature spectrum from three overlapping electrical spectrum analyser (ESA) windows. Each window covered 450 MHz with a resolution bandwidth of 1 MHz. This resolution is small compared with the fitted linewidths, so instrumental broadening from the ESA makes only a small contribution to the measured widths. The three measurements used laser-lock frequencies of 10.450, 10.630 and 10.758 GHz. Each trace was converted to the common Brillouin-frequency axis using Eq. (2.2). Ten background traces were recorded with the Brillouin signal suppressed. The background traces are converted to linear power and averaged before being subtracted from the signal traces. Small differences in power between acquisitions are corrected using the overlap regions so that the corrected spectra can be combined onto a common frequency grid. Small relative frequency offsets were also visible in the overlap regions. We do not correct the frequency axes because there is no independent reference for deciding which acquisition is more accurate. Instead, the observed mismatch is included as a systematic contribution to the uncertainty of the fitted mode frequencies and spacings.

A.2 Gain-dependent Brillouin line shape

The linear Lorentzian fit used for the main spectrum is the low-gain limit of the more general stimulated Brillouin response. For the multimode spectrum, we write the frequency-dependent Brillouin gain as a sum of Lorentzian resonances,

$$G(f) = \sum_{m=1}^N \frac{G_m}{1 + 4(f - f_m)^2 / \Gamma_m^2}, \quad (\text{A.1})$$

where G_m is the peak gain of the m th resonance, f_m is its centre frequency and Γ_m is its FWHM linewidth. In the undepleted-pump approximation, the Stokes power is then [5]

$$P_S(f) = P_0 \exp \left[\sum_{m=1}^N \frac{G_m}{1 + 4(f - f_m)^2 / \Gamma_m^2} \right], \quad (\text{A.2})$$

where P_0 is an approximately frequency-independent background level.

In the low-gain limit, $G(f) \ll 1$, we use

$$\exp[G(f)] \simeq 1 + G(f). \quad (\text{A.3})$$

After subtracting the constant background, Eq. (A.2) reduces to

$$P_S(f) - P_0 \propto \sum_{m=1}^N \frac{G_m}{1 + 4(f - f_m)^2 / \Gamma_m^2}, \quad (\text{A.4})$$

which is the form used for the primary spectrum in Chapter 2. The L_{01} resonance shows the largest deviation from this low-gain form.

At higher gain, the exponential response narrows the apparent line shape. Near the centre of a single resonance,

$$\frac{1}{1 + 4(f - f_m)^2/\Gamma_m^2} \simeq 1 - \frac{4(f - f_m)^2}{\Gamma_m^2}. \quad (\text{A.5})$$

Substituting this into Eq. (A.2) gives

$$P_S(f) \propto \exp \left[-\frac{4G_m(f - f_m)^2}{\Gamma_m^2} \right] \quad (\text{A.6})$$

near the resonance maximum. The line shape near resonance therefore approaches a Gaussian as the gain increases, becoming narrower than the low-gain Lorentzian response for $G_m > \ln 2$ [5]. The linear and high-gain forms are therefore two limits of the same SBS response.

A.3 Near-threshold spectrum

We record a second spectrum closer to the SBS threshold and fit it using the exponential sum of Lorentzian gains in Eq. (A.2), rather than the linear low-gain approximation. The spectrum was recorded in a single 480 MHz ESA window, so no stitching between separate acquisitions was required. Its absolute heterodyne offset was not recorded, leaving an unknown constant offset on the frequency axis. This does not affect the relative positions of the modes or their neighbouring spacings. Figure A.1 shows the single-window spectrum together with its exponential fit. The first two neighbouring-mode spacings obtained from

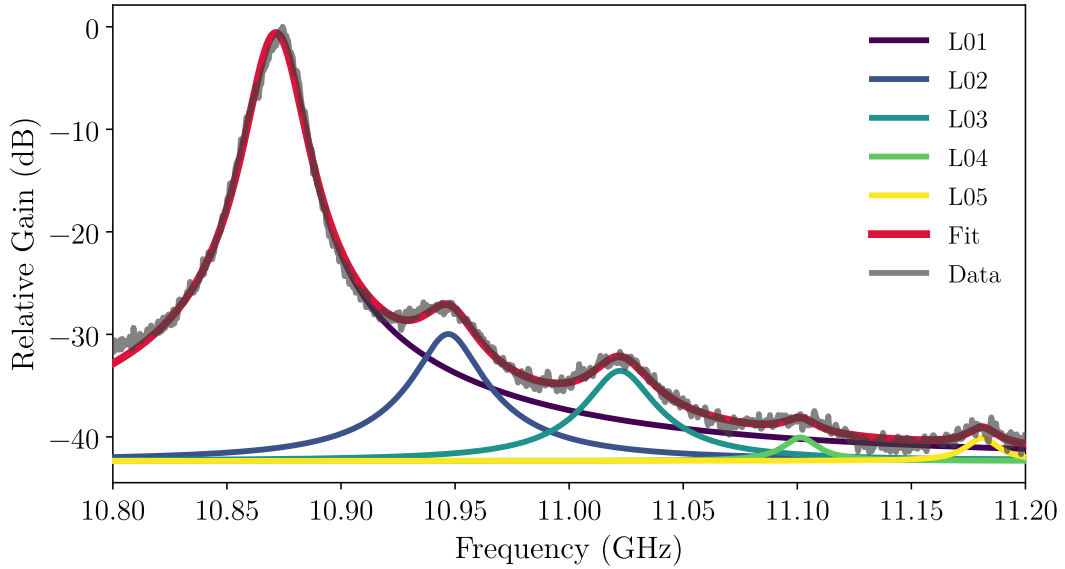


Figure A.1: Near-threshold multimode Brillouin spectrum and fit using the exponential sum of Lorentzian gains in Eq. (A.2). The spectrum was recorded in a single ESA window. Its absolute frequency has an unknown constant offset, so only the relative mode positions are significant.

this spectrum are

$$\Delta f_1 = 75.56 \text{ MHz}, \quad \Delta f_2 = 75.31 \text{ MHz}. \quad (\text{A.7})$$

These are close to the spacing measured in the primary dataset and provide an independent check on the frequency scale of the multimode spectrum.

A.4 Frequency uncertainty

The uncertainties relevant to the later analysis are those of the frequency differences between neighbouring modes. We estimate the statistical contribution by resampling the residuals of the spectral fit, adding them back to the fitted spectrum, and refitting the resulting spectra.

A separate systematic contribution comes from the relative frequency mismatch observed between overlapping ESA windows. We combine the statistical and stitching contributions to give the spacing uncertainties reported in Chapter 2.

The first spacing is additionally sensitive to the choice of fitting model and fit window. Across the fits considered in the analysis, Δf_1 varies by approximately 2 MHz to 3 MHz. This variation is not included in the quoted spacing uncertainty and is treated separately as a fit-model systematic.

The absolute laser-lock uncertainty is largely common to modes within the same acquisition and mostly cancels in neighbouring-mode differences.